

BRIDGE COURSE

GRADE – 8

PHYSICS

Grade 7

Grade 8



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1. ELECTRIC CHARGE AND CURRENT

1.1 Electric Charge

Electric charge can be defined as a fundamental property of subatomic particles that gives rise to the phenomenon of experiencing force in the presence of electric and magnetic fields.

Types of Charge

There are two types of charges:

(a) Positive charge: Positive charges are associated with protons, which are subatomic particles residing in the nucleus of an atom. They are represented by the symbol “+”.

(b) Negative charge: Negative charges are linked to electrons, which orbit the atomic nucleus and are denoted by the symbol “-”.

Unit of Charge

The SI unit of electric charge is coulomb (C).

Small charges are measured in milli-coulomb ($1 \text{ mC} = 10^{-3} \text{ C}$) and micro-coulomb ($1 \mu\text{C} = 10^{-6} \text{ C}$).

Properties of Charge

(a) Charge is a scalar quantity: It has no direction but magnitude only.

(b) Charge is transferable.

(c) Charge resides on the outer surface of a conductor.

(d) Charge is always associated with mass.

(e) Similar charges repel each other while opposite charges attract each other.

(f) Charge is quantized: Electric charge always exists as an integral multiple of the fundamental unit of charge (e).

$e = \text{elementary charge (charge on 1 electron)} = 1.6 \times 10^{-19} \text{ C}$

i.e., $Q = \pm ne$.

Where $n = 1, 2, 3, \dots, \infty$ but $n \neq 2/3, 1/2$ or any fraction.

(g) Charge is conserved: The total charge of an isolated system always remains constant. Charge can neither be created nor be destroyed but can be transferred from one body to another by certain methods like conduction and induction. This is known as "Law of conservation of charge."

1.2 Conductors, Insulators, and Semiconductors

(a) Conductors: The materials through which electric current can flow easily are known as conductors. All the metals like silver, copper, aluminium, etc., are conductors. Carbon, in the form of graphite, is a conductor and the aqueous solutions (water solutions) of salts are also conductors. The human body is a fairly good conductor.

(b) Insulators: The materials which does not allow electric current to flow through them are known as insulators. Rubber, glass, pure water, oil, air, etc. are some examples of insulators.



(c) Semiconductors: The materials whose electrical conductivity lies between conductors and insulators are called semiconductors. Silicon, Germanium etc., are some examples of semiconductors.

1.3 Electric Current

The electric current is a flow of electric charges (called electrons) in the conductor (metal wire). It is the amount of electric charge passing through a conductor in unit time. If a charge of Q coulombs flows through a conductor in time ' t ' second, then the magnitude of the electric current I flowing through it is given by:

$$I = \frac{Q}{t}$$

Where, $Q = ne$,

Here, n = number of electrons, e = charge of electron = $-1.6 \times 10^{-19} \text{ C}$

1C = charge of 6.25×10^{18} electrons.

Unit of Electric Current

The S.I. unit of current is ampere (A). When one coulomb of charge flows through any cross-sectional area of a conductor in 1 second, the electric current flowing through it is said to be 1 ampere.

$$1\text{A} = \frac{1\text{C}}{1\text{s}}$$

Small currents are measured in milli-ampere ($1 \text{ mA} = 10^{-3} \text{ A}$) and micro-ampere ($1 \mu\text{A} = 10^{-6} \text{ A}$).

Flow of Electric Current in a Wire

An electric current is the flow of electrons in a metal wire (or conductor) when a cell or battery is attached across its ends.

A metal wire has plenty of free electrons in it. When the metal wire is not connected to a source of electricity like a cell or a battery, then the electrons present in it move randomly in all directions of the metal wire as shown in the figure.

When a source of electricity like a cell or a battery is connected between the ends of the metal wire, then a force acts on the electrons present in the wire. Since the electrons are negatively charged, they start moving from the negative end to the positive end of the wire. These electrons constitute the electric current in the wire.

Direction of Electric Current

- The direction of electric current is from the positive terminal of the cell to the negative terminal through the circuit.
- Conventionally, in an electric circuit, the direction of electric current is taken as opposite to the direction of the flow of electrons, which are negative charges.

Examples

- (i) **An object has a charge of $-3.2 \times 10^{-19} \text{ C}$. How many excess electrons are present on it? (Charge of one electron, $e = 1.6 \times 10^{-19} \text{ C}$)**

Solution: Given: Objects has Charge (Q) = $-3.2 \times 10^{-19} \text{ C}$

Charge of one electron, $e = 1.6 \times 10^{-19} \text{ C}$

Then, numbers of electrons (n):

$$n = \frac{|Q|}{e} = \frac{3.2 \times 10^{-19}}{1.6 \times 10^{-19}} = 2$$

(ii) **If a body gains 1250 electrons, find the total charge on the body.**

Solution: Given:

No. of Electrons (n) = 1250

Charge on one electron = $-1.6 \times 10^{-19}\text{C}$

Then, Total Charge (Q):

$$Q = ne = 1250 \times (-1.6 \times 10^{-19}) = -2.0 \times 10^{-16}\text{C}$$

(iii) **Classify the following materials as conductor, insulator, or semiconductor:**

Copper, Rubber, Silicon, Human body, Pure water.

Answer: Copper $\rightarrow$ Conductor

Rubber $\rightarrow$ Insulator

Silicon $\rightarrow$ Semiconductor

Human body $\rightarrow$ Conductor

Pure water $\rightarrow$ Insulator

(iv) **In a metal wire, electrons move from the negative terminal to the positive terminal. What is the direction of conventional current?**

Answer: Conventional current is defined as the direction of flow of positive charge.

Therefore, it flows from the positive terminal to the negative terminal.

(v) **If a conductor wire of 15 C of charge is passed for 3 sec, then find electric current.**

Solution: Given: Charge, $Q = 15\text{ C}$ and Time, $t = 5\text{ sec}$

$$I = \frac{Q}{t} = \frac{15}{3} = 5\text{ A}$$

(vi) **Find the amount of electric charge flowing through the circuit if an electric current of 7 A is drawn by an electric appliance for 3 minutes.**

Solution: Given: Current, $I = 7\text{ A}$, Time, $t = 3\text{ minutes} = 60 \times 3 = 180\text{sec}$

We know that,

$$\text{Current, } I = \frac{Q}{t}$$

$$\text{Therefore, } 7 = \frac{Q}{180}$$

$$Q = 180 \times 7\text{C}$$

$$Q = 1260\text{C}$$



Practice Questions - 1

- (i) 1 Coulomb charge is equal to:
 - (A) Charge of 1 electron
 - (B) Charge of 6.25×10^{18} electrons
 - (C) Charge of 10^{18} electrons
 - (D) Charge of 100 electrons
- (ii) In a metal wire connected to a battery, electrons move:
 - (A) From positive terminal to negative terminal
 - (B) From negative terminal to positive terminal
 - (C) Randomly even when battery is connected
 - (D) Only inside the battery
- (iii) The SI unit of electric current is:
 - (A) Coulomb
 - (B) Ampere
 - (C) Volt
 - (D) Watt
- (iv) Which of the following materials is a semiconductor?
 - (A) Copper
 - (B) Rubber
 - (C) Silicon
 - (D) Glass
- (v) Which of the following is a good conductor of electricity?
 - (A) Pure water
 - (B) Glass
 - (C) Graphite
 - (D) Rubber
- (vi) If 1 coulomb of charge flows through a conductor in 2 seconds, the current is:
 - (A) 2 A
 - (B) 1 A
 - (C) 0.5 A
 - (D) 0.2 A
- (vii) A charged body has 4.8×10^{-19} C charge. Find the number of electrons transferred.
- (viii) State any two properties of electric charge.
- (ix) Differentiate between conductors and insulators. Give one example of each.
- (x) 4×10^{20} electrons are flowing through a conducting wire in 16 seconds. The amount of current flowing through the conducting wire is 2k Amperes, then find out the value of k.

2. ELECTRIC CIRCUIT AND HEATING EFFECT

2.1 Electrical Cells and Battery

Electric Cells

An electric cell is a device, which converts chemical energy into electrical energy (electric current).

It mainly consists of two components:

(a) Electrolyte: This is a chemical component that conducts electricity.

(b) Electrodes: There are two electrodes in a cell. One is positive electrode called the anode and the other is negative electrode called cathode. When electrodes come in contact with electrolyte, a chemical reaction takes place inside the cell to produce electricity.

Battery

A Battery is formed when we join two or more than two cells in such a way that negative terminal of one cell is connected to positive terminal of another cell and so on.



For example: A battery combination of four cells is given in the adjacent diagram.

Types of Cells

Depending on the continuity of the flow of electric current, cells are classified into two types. They are:

- (a) Primary cells
- (b) Secondary cells

(a) Primary Cells: The cells which cannot be charged again and again are known as primary cells. They are used in clocks, watches, toys, etc. Therefore, the primary cells are usually produced in small sizes.

Example: Voltaic, Daniel, Leclanche cell, and Dry cells.

(b) Secondary Cell: Cells which are reusable and rechargeable are known as secondary cells. The size of the secondary cells can be small or even large, depending upon the usage. Mobile phone consumes less charge, so the secondary cells used in mobiles are the size of a palm, Whereas the cells used in automobiles like cars and buses are large and very heavy because they consume more current.

Examples: Edison cells, lead-acid accumulators and Nickel-Iron accumulators.

2.2 Circuit Diagram

The diagrams which show the arrangement of electric components in a circuit are called circuit diagrams.

A circuit diagram is a symbolic representation of an electric circuit. It shows how the electric components are connected together. Electricians and engineers draw circuit diagrams which help them to design the actual circuits.

A circuit generally has:

- a source of electric current- a cell or battery.
- a device which consumes the electricity- a bulb, etc.
- a key or a switch- this may be connected anywhere along the circuit to stop or allow the flow of current.
- connecting wires for carrying current.

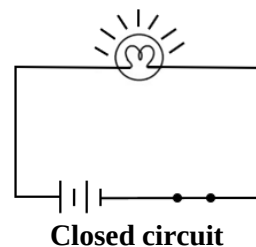
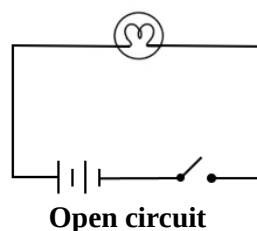
2.3 Electric Circuit

The path of the flow of electricity starting from one terminal of the cell and returning to the other terminal is called an electric circuit.

Types of Electric Circuit

There are two types of electric circuits:

(a) Open Electric Circuit: An electric circuit through which electric current can't flow is known as an open electric circuit. when the switch is in the 'OFF' position,



(b) Closed Electric Circuit: An electric circuit through which electric current can flow continuously is known as a closed circuit. When the switch is in the 'ON' position, the circuit from the positive terminal of the battery to the negative terminal is complete and hence, current will flow.

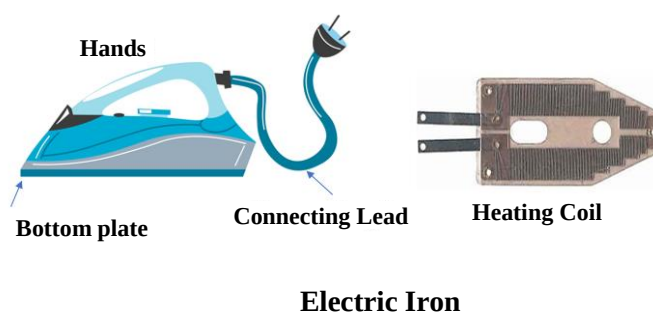
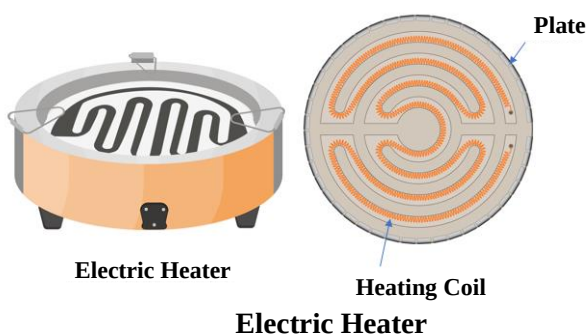
2.4 Heating Effect of Electric Current

When an electric current is passed through a conductor, it generates heat due to the hindrance caused by the conductor to the flowing current. This effect is called **the heating effect of electric current**.

The heat produced in a wire depends on its material, length and thickness. Thus, for different requirements, wires of different materials, lengths and thicknesses are used.

Heating Effect of Electric Current is Used in Various Appliances like:

- Electric Bulb where electric current is passed through a tungsten filament which glows and produces light and heat.
- Electric Iron and Electric Heater where filament is made up of nichrome wire as a coil that gets heated up when electric current flows through it.



Examples

(i) **How is a battery formed? Draw a simple representation.**

Answer: A battery is formed by connecting two or more cells together.

The negative terminal of one cell is connected to the positive terminal of the next cell. This arrangement increases the total voltage.

Representation:



Cells connected in series form a battery.

(ii) **What is a circuit diagram? Name the basic components shown in it.**

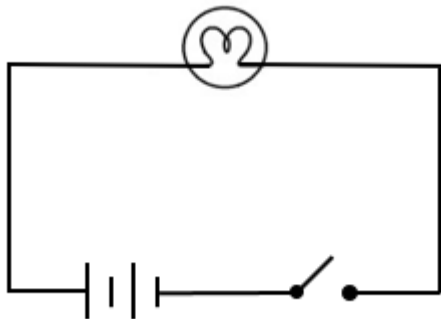
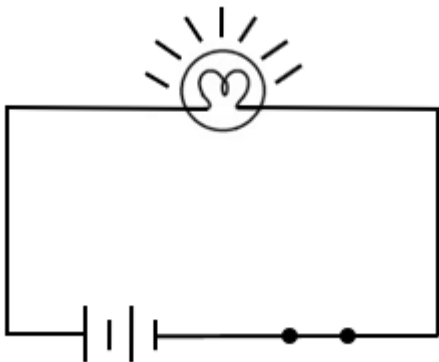
Answer: A circuit diagram is a symbolic representation of an electric circuit.

It shows how different components are connected using standard symbols.

Basic components:

- Source of current (cell or battery)
- Bulb or device (load)
- Switch (key)
- Connecting wires

(iii) **What is the difference between open circuit and closed circuit?**

Open Circuit	Closed Circuit
(a) An electric circuit through which electric current cannot flow is known as open circuit.	(a) An electric circuit through which electric current can flow continuously is known as closed circuit.
(b) 	(b) 
(c) Electrical devices cannot function in it.	(c) Electrical devices can function.

(iv) **Name some devices which work on heating effect of electric current.**

Answer: There are some Devices which work on the principle of heating effect of electric current. Electric bulb, Electric iron, Electric geyser and Electric fuse, etc.

(v) **Why is nichrome used in electric heaters instead of copper?**

Answer: Nichrome has high resistance and produces more heat when current flows through it. Copper has low resistance and produces less heat. Therefore, nichrome is used in heaters and irons.



Practice Questions - 2

- (i) An electric cell converts:
 - (A) Electrical energy into chemical energy
 - (B) Chemical energy into electrical energy
 - (C) Heat energy into electrical energy
 - (D) Light energy into chemical energy
- (ii) A bulb does not glow in a circuit. The most probable reason is:
 - (A) Circuit is closed
 - (B) Battery is connected properly
 - (C) Switch is ON
 - (D) Circuit is open
- (iii) Which of the following is a secondary cell?
 - (A) Dry cell
 - (B) Leclanche cell
 - (C) Lead-acid accumulator
 - (D) Voltaic cell
- (iv) Current cannot flow in a circuit when the switch is:
 - (A) ON
 - (B) Closed
 - (C) OFF
 - (D) Connected
- (v) Nichrome wire is used in heaters because it:
 - (A) Melts easily
 - (B) Produces heat when current flows
 - (C) Does not conduct electricity
 - (D) Is very soft
- (vi) The heat produced in a wire increases when:
 - (A) The wire is thicker
 - (B) The length of wire is reduced
 - (C) The resistance of wire is higher
 - (D) The wire is replaced by plastic
- (vii) Differentiate between primary and secondary cells.
- (viii) What is an electric cell? Name its two main components.
- (ix) Explain the heating effect of electric current. Also write any three applications of this effect in daily life.
- (x) Draw a labelled diagram of:
 - (a) Open electric circuit
 - (b) Closed electric circuitExplain both the circuit with real life examples.

3. MAGNETS

Objects which attract magnetic materials like cobalt, nickel, iron, etc, are called magnets.

3.1 Magnetic and Non-Magnetic Material

Magnetic Materials

The materials that are attracted to a magnet are called magnetic materials. Iron, nickel, cobalt, etc, are called magnetic materials. Magnetite or Lodestone is the first known natural magnet which contains iron.



Cobalt



Iron



Nickel

Non-Magnetic Materials

The materials that are not attracted to a magnet are called non-magnetic materials. For example, leather, paper, plastic, etc.



Leather



Paper



Plastics

3.2 Types of Magnets



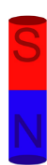
Horseshoe Magnet



U-Shaped Magnet



Bar Magnet



Cylindrical Magnet



Needle Magnet



Circular Magnet



Ring Magnet



Rod Magnet



Oval shaped Magnet

Different Shapes of Magnets

3.3 Poles of a Bar Magnet

The ends of a bar magnet that attracts more iron filings are called poles.

All magnets have two poles- North (N) and South (S) pole.

When a bar magnet is placed on a sheet spread with iron filings, they get attracted towards the ends of the bar magnet.

Note: The magnetic force is maximum at the poles of a magnet.



Poles of Bar Magnet

3.4 Properties of Magnets

(a) Attractive property

A magnet has the property to attract some substances like iron, cobalt, nickel etc when brought near it.

This property of magnet is called its attractive property.



Bar Magnets attracting iron filings at the poles

(b) Directive property

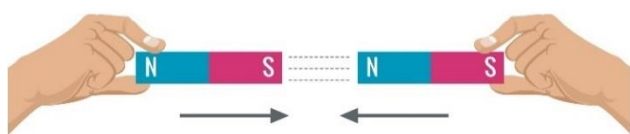
If we suspend a magnet freely with a thread then it always aligns itself in the geographic north-south direction when it comes to rest. This is called the directive property of a magnet.

(c) Each magnet has two poles which exist in pairs.

Magnetic monopoles do not exist separately. If a magnet is cut into two pieces each piece will behave like an independent magnet, with a north pole and a south pole.

(d) Like poles repel each other and unlike poles attract each other.

- When two magnets are moved close to each other, they either attract or repel each other.
- Attraction or repulsion of the magnets depends on the direction of the poles facing each other.
- The opposite poles (N-S or S-N) of magnets attract, whereas similar poles (N-N or S-S) of magnets repel when they are brought near each other.



Attraction



Repulsion

3.5 Electromagnets

The coils which behave as a magnet when electric current flows through it and loses its magnetic property when the electric current is switched off, such coils are known as **electromagnets**.

Uses of Electromagnet:

- The electromagnets can be made very strong and can lift very heavy loads.
- The electromagnets can also be used to separate magnetic material from a junkyard.
- Doctors use tiny electromagnets to take out small pieces of magnetic materials that have accidentally fallen in the eye.
- Many toys also have electromagnets inside them.

Factors which affect the strength of electromagnets:

- It depends on the number of turns in the coil.
- It depends on the magnitude of current passing through the conductor.
- It depends on the nature of the core inside the coil.

Examples

(i) **What are non-magnetic materials?**

Answer: The materials that are not attracted to a magnet are called non-magnetic materials.
Example: Wood, plastic etc.

(ii) **How to determine the north pole of a bar magnet?**

Answer: A freely suspended bar magnet always points in a north-south direction. The end that points towards the geographic north direction is the north pole of the magnet while the end that points towards the south direction is the south pole of the magnet.

(iii) **Which property of a magnet is used for the testing of a permanent magnet?**

Answer: We use repulsion property for testing a magnet. A permanent magnet causes repulsion with one pole when the poles are brought near a suspended magnet.

(iv) **A magnet is broken into 4 pieces. How many pair of poles we will get?**

Answer: When you break the magnet into 4 pieces, each piece will have 2 poles (a north and a south). Since each pair consists of one north pole and one south pole, the total number of pairs of poles is 4.

(v) **A straight iron nail is wrapped with insulated copper wire. When current flows through the coil, the end A behaves as a North pole. What will be the pole at the other end B?**

Answer: An electromagnet always has two poles — North and South.
If one end is North, the other end must be South.



Practice Questions - 3

- (i) Which of the following materials is considered as magnetic?
 - (A) Plastic
 - (B) Cobalt
 - (C) Lead
 - (D) Wood
- (ii) What happens when a magnet is brought near a non-magnetic material?
 - (A) The material is attracted to the magnet
 - (B) The material is repelled by the magnet
 - (C) The magnet has no effect on the material
 - (D) The material becomes magnetized
- (iii) What happens when a magnet is cut into two pieces?
 - (A) One piece will lose its magnetic properties
 - (B) One piece will have North pole and another pole will have South pole
 - (C) Each piece will behave like an independent magnet with both a North and South pole
 - (D) Only one piece will retain its magnetic properties
- (iv) If the current through a coil is doubled, the electromagnet will become:
 - (A) Stronger
 - (B) Demagnetised
 - (C) Weaker
 - (D) Permanent
- (v) Which of the following does NOT affect the strength of an electromagnet?
 - (A) Number of turns of coil
 - (B) Current passing through coil
 - (C) Nature of core
 - (D) Colour of wire
- (vi) What happens to an electromagnet when the electric current supply is turned off?
 - (A) It becomes a permanent magnet.
 - (B) It retains its magnetic properties.
 - (C) It loses its magnetic properties.
 - (D) It heats up.
- (vii) What is directive property of magnet?
- (viii) Differentiate between magnetic and non-magnetic materials.
- (ix) List the properties of the poles of a magnet. When a magnet is cut into 3 equal parts how many total pairs of poles they have?
- (x) Define electromagnet. Explain its construction and write any three uses.

4. LIGHT

Light is a form of energy which helps us see objects.

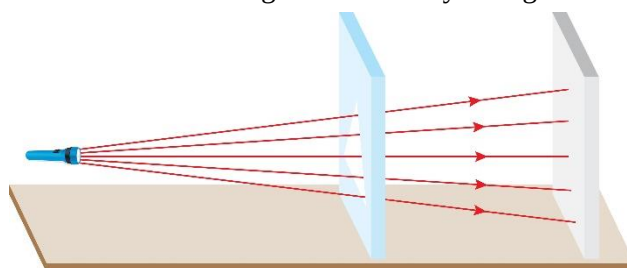
4.1 Properties of Light

- Light is an electromagnetic wave and does not require any material medium to travel.
- Light travels in a straight line with a speed of 3×10^8 m/s (in vacuum/air). It propagate in different media with different speeds, the maximum being in air or vacuum.
- Light has dual nature i.e., wave as well as particle.
- Light casts a shadow when it falls on an obstacle.

4.2 Ray and Beam of Light

Ray of Light

A straight line drawn in the direction of motion of light is called Ray of Light.

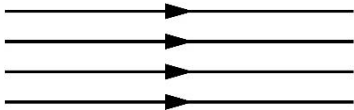
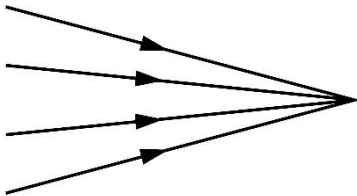
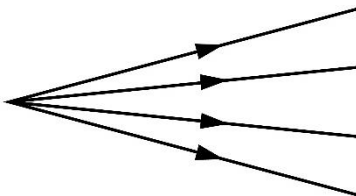


Ray of Light

Beam of Light

A bunch of light rays or bundle of light rays is called a Beam of Light.

Different Types of Beam of Light

Parallel Beam of Light	Convergent Beam of Light	Divergent Beam of Light
These rays of light are equidistant from each other at all places and do not meet. 	These rays of light come from different directions and meet or appear to meet at a point. 	These rays of light are emitted from a common source (point) of light and travel in different directions after emitting. 
Parallel beam of light	Convergent beam of light	Divergent beam of light

4.3 Sources of Light

Classification on the basis of light-emitting ability

(a) Luminous Objects: The objects which emit their own light are known as luminous objects. These objects may be natural or man-made.

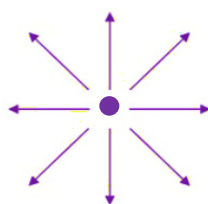
For example: Sun, fireflies, torch, bulb, etc.

(b) Non-luminous Objects: The objects which does not emit their own light are known as non-luminous objects. However, such objects become visible due to the reflection of the light falling on them.

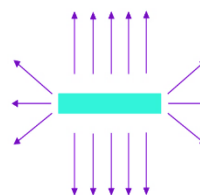
For example: Moon, table, chair, etc.

Classification on the basis of the size of the object

(a) Point source of light: A source of light that is of the size of a pin head is called a Point source of light.



A Point Source of light



An extended source of light

4.4 Rectilinear Propagation of Light

If we light a torch on a dark night, we will see that the beam of light produced, travels in straight line.

The property of light travelling in a straight line is called Rectilinear Propagation of Light.

Examples:

- The beams of searchlight at the airport shows that light travels in a straight line.
- The beam of light coming from the projection room of a cinema hall and falling on the screen also shows that light travels in a straight line.



Light travels in a straight line

4.5 Transparent, Translucent and Opaque Objects

We can classify materials into three categories according to their optical properties:

Transparent Objects	Translucent Objects	Opaque Objects
These objects allow light to pass through them completely.	These objects allow light to pass through them partially.	These objects do not allow light to pass through them.
We can clearly see through transparent objects.	We can see through translucent objects, but not very clearly.	We cannot see through opaque objects.
Transparent objects do not cast shadows because they do not block light.	Translucent objects cast faint shadows as they block light partially.	Opaque objects cast dark shadows as they block light completely.

Substance	See-Through	Shadow Cast	Classification
Pencil	Not at all	Dark	Opaque
Paper	Not at all	Dark	Opaque
Glass	Fully	No shadow	Transparent
Water	Fully	No shadow	Transparent
Smoke	Partially	Faint	Translucent
Butter Paper	Partially	Faint	Translucent

4.6 Reflection of Light

When a ray of light falls upon any smooth surface, it is sent back into the same medium from which it came. This phenomenon is called Reflection of light.

Laws of Reflection

(a) At the point of incidence, the incident ray, the reflected ray and the normal, all lie in the same plane.

(b) Angle of incidence is equal to the angle of reflection
i.e., $\angle i = \angle r$.

Note:

- Laws of reflection apply to the plane surface as well as curved surfaces.
- When a ray of light falls on a mirror normally or at a right angle to the surface, it gets reflected along the same path.

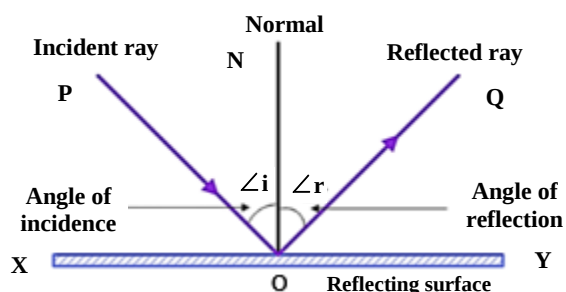


Fig. 4.16 Reflection of light

Examples

(i) A Wooden piece is an _____ object.

Answer: Opaque.

(ii) The objects which partially allow light to pass through them are called _____.

Answer: Translucent objects.

(iii) Differentiate between a ray of light and a beam of light.

Ray of Light	Beam of Light
A straight line drawn in the direction of motion of light	A collection of parallel or diverging ray.
Represents the path of light	Can consist of multiple rays moving together in same direction.

(iv) Explain why the beam of light coming from a cinema projector appears straight.

Answer: Light travels in a straight line. This property is called rectilinear propagation of light. The straight beam from the projector demonstrates this property.

(v) **What happens to a light ray that is incident on a plane mirror normally?**

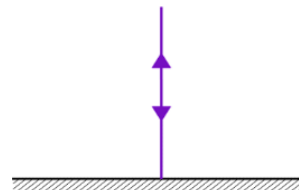
Answer: When a light ray is incident on a plane mirror normally,

The angle of incidence, $\angle i = 0$

According to the laws of reflection, $\angle i = \angle r$

So, the angle of reflection also must be zero.

The reflected ray must travel along the same path in the opposite direction as shown in the adjacent figure.



(vi) **If a light ray incident on a plane mirror makes an angle of 30° with it, find the angle of incidence and angle of reflection.**

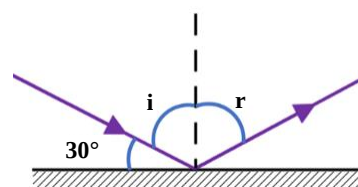
Answer: It is given that the incident ray makes an angle of 30° with the mirror.

But the angle of incidence is the angle between the incident ray and the normal.

Here, $\angle i + 30^\circ = 90^\circ$

So, $\angle i = 90^\circ - 30^\circ = 60^\circ$

On applying laws of reflection, $\angle i = \angle r = 60^\circ$





Practice Questions - 4

- (i) A bunch of light rays or bundle of light rays is called:
 - (A) ray of light
 - (B) beam of light
 - (C) both (A) and (B)
 - (D) none of these
- (ii) Light shadow of the object will be formed when the object is:
 - (A) Transparent
 - (B) Translucent
 - (C) Opaque
 - (D) None of these
- (iii) For what angle of incidence do the normal, the incident ray, and the reflected ray all coincide?
 - (A) 0°
 - (B) 45°
 - (C) 90°
 - (D) 30°
- (iv) How does light travel?
 - (A) Rectilinear propagation
 - (B) Random motion
 - (C) Curvilinear motion
 - (D) Circular motion
- (v) Which of the following are non-luminous bodies?
 - (i) A planet
 - (ii) A star
 - (iii) A metal strip
 - (iv) A burning candle
 - (A) Only (i) and (ii)
 - (B) Only (ii) and (iii)
 - (C) Only (i) and (iii)
 - (D) Only (ii) and (iv)
- (vi) The angle of reflection is the angle between:
 - (A) the incident ray and the surface of the mirror.
 - (B) the reflected ray and the surface of the mirror.
 - (C) the normal to the surface and the incident ray.
 - (D) the normal to the surface and the reflected ray.
- (vii) A ray of light is incident on a plane mirror such that the angle between the incident ray and the reflected ray is 100° . Find the angle of incidence.
- (viii) A ray of light strikes a plane mirror making an angle of 25° with the mirror surface. Find the angle between the incident ray and reflected ray.
- (ix) What are the different kinds of object based on their optical properties?
- (x) What is the Reflection of Light, write the Laws of Reflection and Terms Related to Reflection.

5. MIRRORS

5.1 Image

Real Image: When reflected light rays actually meet at a point after reflection from a mirror respectively, then a real image is formed. A real image forms when reflected light rays converge at a point.

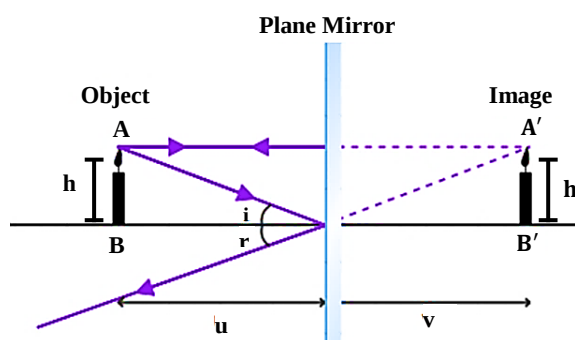
Virtual Image: When reflected light rays appear to meet at a point after reflection from a mirror respectively, then a virtual image is formed. A virtual image forms when reflected light rays diverge from a point.

5.2 Plane Mirror

A plane mirror is a mirror with a flat (planar) reflective surface. For light rays striking a plane mirror, the angle of incidence equals the angle of reflection.

Characteristics of Image formed by a Plane Mirror

- The size of the image is same as the size of the object. i.e. $h = h'$
- The distance of the object in front of the mirror is equal to the distance of the image behind the mirror. i.e. $u = v$
- A virtual and erect image is formed.
- The image formed by a plane mirror is laterally inverted.



Note:

The magnitude of velocity of the image formed by a plane mirror is the same as the magnitude of velocity of the object when the mirror is at rest. The direction of image is opposite to the direction of velocity of object i.e., $(\mathbf{v}_o = -\mathbf{v}_i)$

5.3 Spherical Mirrors

A mirror whose polished and reflecting surface is part of a hollow glass sphere is called a **spherical mirror**

There are two types of spherical mirrors:

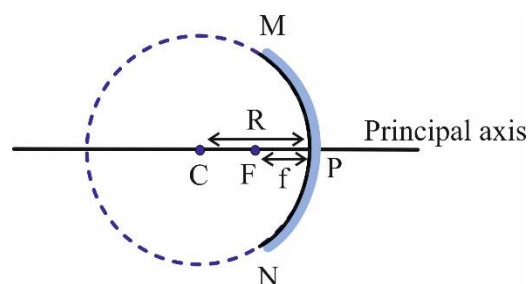
(a) Concave (Converging) Mirror: A spherical mirror whose inner hollow surface is the reflecting surface.

(b) Convex (Diverging) Mirror: A spherical mirror whose outer bulging surface is the reflecting surface.

Focal length (f): The distance between the pole and the focus is called the focal length. The focal length of concave mirror is negative, while it is positive for convex mirror.

For a spherical mirror, the radius of curvature and focal length are related as:

$$f = \frac{R}{2}$$



Important points to note about the Images formed by a Spherical Mirror

- Only a concave mirror forms a real image of a real object.
- Virtual images can be formed by all mirrors, i.e., plane, concave and convex. Plane mirrors always form a virtual image of the same size as that of the object. Virtual images formed by concave mirrors are always magnified. Convex mirrors always form a diminished virtual image.

Uses of Concave and Convex Mirrors

Concave Mirror		Convex Mirror
(a)	Concave mirrors are used in torches, searchlights, and vehicle headlights to get a powerful parallel beam of light (the light source is placed at focus).	Convex mirrors are used as rear-view mirrors in vehicles because a full-length image of a vehicle behind you can be seen in a small convex mirror. (i) they always give an erect image. (ii) they have a wider field of view as they are curved outwards.
(b)	Concave mirrors are used by dentists to see large images of patient's teeth.	Convex mirror is used in shops as a security mirror.
(c)	A concave mirror is used as a shaving mirror to see an enlarged image of the face.	Convex mirrors are used at blind turns and on points of merging traffic to facilitate the vision of both sides of traffic.
(d)	Concave mirrors are used to concentrate sunlight to produce heat in a solar cooker.	Convex mirror is utilized in ATMs and other places for security reasons.

5.4 Reflection of Light

The phenomenon of bending of light when it enters obliquely from one medium to another medium is called the refraction of light. It is caused due to a change in the velocity of light while travelling from one medium to another.

The speed of light is maximum in vacuum and is equal to 3×10^8 m/s.

Some examples of refraction:

(a) The bottom of the swimming pool appears nearer.



(b) A partially immersed pencil in water appears to be bent at the interface of water and air.



(c) Lemons placed in a glass tumbler appear bigger.



(d) Letters of a book appear to be raised and magnified when seen through a convex lens.



Examples

- (i) A boy standing in front of a plane mirror finds his image to be at a distance of 10 m from himself. The distance of his image from the mirror is ____m.

Solution: Distance between the boy and his image = 10 m

Distance between boy and mirror + distance between mirror and image = 10 m

But the distance between boy and mirror (object distance) = distance between mirror and image

Therefore, the distance of his image from the mirror = $\frac{10}{2} = 5$ m

(ii) **Differentiate between a real image and a virtual image.**

Answer:	Real Image	Virtual Image
	A real image is formed when reflected rays actually meet at a point after reflection. Real images are generally inverted.	A virtual image is formed when reflected rays appear to meet at a point. Virtual images are erect.
	Real image is formed by actual convergence of reflected rays.	Virtual image is formed by apparent convergence of reflected rays.

(iii) **A person stands 4 m in front of a plane mirror. Find:**

(a) Distance of image from the mirror

(b) Distance between the person and the image

Solution: (a) For a plane mirror:

Object distance (u) = Image distance (v)

Given: $u = 4\text{m}$

Therefore, $v = 4\text{m}$

(b) Distance between object and image:

$= u + v = 4 + 4 = 8\text{m}$

(iv) **A boy moves towards a plane mirror with a speed of 3 m/s. Find the speed and direction of his image.**

Solution: For a plane mirror:

Magnitude of velocity of image = magnitude of velocity of object

Direction is opposite

Given: $v_o = 3\text{ m/s}$

Therefore: $v_i = -3\text{ m/s}$

The image moves with speed 3 m/s in the opposite direction.

(v) **The radius of curvature of a spherical mirror is 40 cm. Find its focal length.**

Solution: Formula:

$$f = \frac{R}{2}$$

$$f = \frac{40}{2} = 20\text{ cm}$$

If mirror is concave $\rightarrow f = -20\text{ cm}$

If mirror is convex $\rightarrow f = +20\text{ cm}$

Focal length = 20 cm (sign depends on type of mirror).



Practice Questions - 5

- (i) Which of the following statements correctly defines a "Real Image"?
 - (A) It forms when reflected light rays diverge from a point.
 - (B) It forms when reflected light rays actually meet at a point.
 - (C) It is always formed by a convex mirror.
 - (D) It is always erect
- (ii) If the radius of curvature (R) of a spherical mirror is 40 cm, what is its focal length (f)?
 - (A) 80 cm
 - (B) 20 cm
 - (C) 40 cm
 - (D) 10 cm
- (iii) Which mirror is specifically chosen as a rear-view mirror in vehicles because it provides a wider field of view?
 - (A) Concave Mirror
 - (B) Plane Mirror
 - (C) Convex Mirror
 - (D) Shaving Mirror
- (iv) What type of image is formed by a plane mirror?
 - (A) Real and Inverted
 - (B) Virtual and Erect
 - (C) Real and Erect
 - (D) Virtual and Inverted
- (v) In a plane mirror, the image appears to be:
 - (A) Larger than the object
 - (B) Smaller than the object
 - (C) Same size as the object
 - (D) Variable in size
- (vi) An object moves towards a plane mirror with a velocity v . What is the magnitude of the velocity of the image?
 - (A) $2v$
 - (B) $v/2$
 - (C) v
 - (D) 0
- (vii) A concave mirror has a radius of curvature of 60 cm. What is its focal length?
- (viii) A man runs towards a stationary mirror at a speed of 15 m/s. What is the speed of his image with respect to the mirror?
- (ix) Define the phenomenon of Refraction of Light. State the primary cause of this phenomenon and provide one example from daily life.
- (x) Compare Concave and Convex mirrors based on their uses and image characteristics.
 - (a) Why are Concave mirrors used in torches and by dentists?
 - (b) Why are Convex mirrors preferred for security in shops and ATMs?
 - (c) Which of these two mirrors always forms a diminished virtual image?

6. VELOCITY AND ACCELERATION

6.1 Distance and Displacement

Distance:

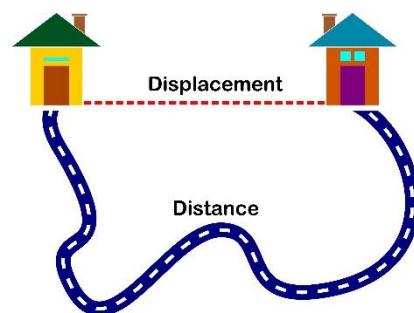
Distance travelled by a body can be defined as the length of the path covered by a body irrespective of the direction of motion.

- It is a scalar quantity.
- The SI unit of distance travelled is metre (m) while the CGS unit is centimetre (cm).

Displacement:

Displacement of a body can be defined as the shortest distance travelled in the direction of motion from the initial to the final point. It is a vector quantity.

Note: If a body travels in such a way that it comes back to its starting point, then the displacement is zero. However, the distance travelled is never zero.



6.2 Speed

Speed: The distance covered by a body in unit time is called 'speed'.

$$\text{Speed} = \frac{\text{Distance travelled}}{\text{Time taken}}$$

Average Speed: It is the total distance travelled by a body divided by the total time taken to cover that distance.

$$\text{Average speed} = \frac{\text{Total distance travelled}}{\text{Total time taken}}$$

6.3 Velocity

The rate of change of displacement with respect to time is called velocity. Mathematically,

$$\text{Velocity} = \frac{\text{Displacement}}{\text{Time taken}}$$

The SI unit of velocity is m/s while the CGS unit is cm/s.

Types of Velocity

(a) Uniform Velocity

When a body moves with uniform speed in a specified direction, it is said to be moving with uniform velocity. Thus, a body is said to be moving with a uniform velocity when its magnitude as well as its direction remains the same.

Examples: Aeroplane moving with a speed of 500 km/hr towards the East.

(b) Non-Uniform Velocity

When a body moves such that either its magnitude or direction or both changes, then it is said to be moving with Non-Uniform velocity.

Examples: A car moving on a straight road such that its speed changes from time to time has Non-Uniform velocity. A car taking a turn has variable velocity as its direction changes.

(c) Average Velocity

When a body moves with Non-Uniform velocity, the average velocity of the body is equal to the ratio of the final displacement of the body to the total time taken.

$$\text{Average velocity} = \frac{\text{Final displacement}}{\text{Total time taken}}$$

Difference between Speed and Velocity

Speed

- (a) It is a scalar quantity.
- (b) $\text{Speed} = \frac{\text{distance travelled}}{\text{time}}$
- (c) It is the rate of change of position of an object.
- (d) It may be positive or zero.

Velocity

- (a) It is a vector quantity.
- (b) $\text{Velocity} = \frac{\text{displacement}}{\text{time}}$
- (c) It is the rate of change of position of an object in a specific direction.
- (d) It may be positive, zero or even negative.

6.4 Acceleration

Acceleration is the rate of change of the velocity of a body with respect to time.

Mathematically,

$$\text{Acceleration} = \frac{\text{Change in velocity}}{\text{Time taken}}$$

$$a = \frac{v - u}{t}$$

Here v = final velocity, u = initial velocity and t = time taken.

The SI unit of acceleration is m/s^2 while the CGS unit is cm/s^2 .

Types of Acceleration

(a) Positive Acceleration

If the velocity of a body increases in the same direction, then the body is said to have a positive acceleration.

Example: A freely falling body.

(b) Negative Acceleration (Retardation/Deacceleration)

If the velocity of a body decreases then, the body has a negative acceleration or it is said to be undergoing retardation or deacceleration.

Example: A train slows down when approaching a station.

(c) Zero acceleration:

When the velocity of the object remains the same throughout a journey of motion then the acceleration is said to be zero, as there is no change in the velocity with the change in time.

Example: A car moving with constant velocity on the road has zero acceleration.

Examples

- (i) A train starts from station A and moves along a straight path towards station B which is 100 km away from A. Reaching B it turns back and stops midway between A and B. Find the distance and displacement of the train. Neglect the length of the train.



Solution: Let's say the midpoint is C.

Therefore, Distance = AB + BC = 100 + 50 = 150 km

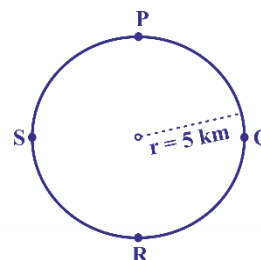
Displacement = AC = 50 km or 50 km in the direction from station A to station B.

- (ii) A motorcyclist rides from point P to Q to R to S and finally to P in a circular path as shown in the adjacent figure.

Find:

(a) the distance travelled by motorcycle.

(b) the displacement.



Solution: Given radius, $r = 5$ km

(a) Here, the distance travelled by motorcycle = circumference of the circle

($\because$ the motorcycle moves one complete rotation)

the distance travelled by motorcycle = $2\pi r$

the distance travelled by motorcycle = $2 \times 3.14 \times 5$

the distance travelled by motorcycle = 31.4 km

(b) Here, the initial point is P and the final point is also P.

Therefore, there will be no change in position and hence the displacement is equal to zero.

- (iii) The distance between two stations is 200 km. A train takes 4 hours to cover this distance. Calculate the speed of the train.

Solution: Given:

Distance = 200 km, Time taken = 4 hours

$$\text{Speed} = \frac{\text{Distance covered}}{\text{Time taken}} = \frac{200}{4} \text{ km/h}$$

Speed = 50 km/h

- (iv) A bus travels 40 km east in 1 hour and 60 km east in next 2 hours. Find average velocity.

Solution: Given:

Total displacement = 40 + 60 = 100 km east

Total time = 1 + 2 = 3 h

$$V_{\text{avg}} = \frac{\text{Total Displacement}}{\text{Total Time}} = \frac{100}{3} = 33.33 \text{ m/s}$$

- (v) The velocity of a car at 10:50 am is 60 km/h and at 10:52 am it is 80 km/h. Assuming constant acceleration in the given period, find its value.

Solution: Given:

Final velocity $v = 80 \text{ km/h}$

Initial velocity $u = 60 \text{ km/h}$

Time taken $t = 2 \text{ min.}$

$$a = \frac{v - u}{t} = \frac{80 \text{ km/h} - 60 \text{ km/h}}{2 \text{ min}} = \frac{20 \text{ km/h}}{(1/30)\text{h}} = 600 \text{ km/h}^2$$

- (vi) A car starts from rest and acquire a velocity of 54 km/h in 2 sec. Find the acceleration by car assume motion of car is uniform?

Solution: Given that, $u = 0$, $v = 54 \text{ km/h} = 15 \text{ m/s}$, $t = 2 \text{ sec}$

Acceleration is given by

$$a = \frac{v - u}{t}$$

$$a = \frac{15 - 0}{2}$$

$$\text{So, } a = 7.5 \text{ m/s}^2$$



Practice Questions - 6

- (i) Which of the following is a vector quantity?
 - (A) Mass
 - (B) Speed
 - (C) Temperature
 - (D) Displacement
- (ii) Which physical quantity does not have direction?
 - (A) Velocity
 - (B) Acceleration
 - (C) Distance
 - (D) Force
- (iii) Which of these motions is periodic?
 - (A) A car on a straight road
 - (B) A bouncing ball
 - (C) A pendulum swing
 - (D) A person walking
- (iv) When does displacement equal distance?
 - (A) Always
 - (B) When object moves in a straight line without changing direction
 - (C) When motion is circular
 - (D) When the object returns to starting point
- (v) A car starts from rest and acquires a velocity of 54 km/h in 2 sec. Find the acceleration.
 - (A) 7.5 m/sec^2
 - (B) 15 m/sec^2
 - (C) 5 m/sec^2
 - (D) 10 m/sec^2
- (vi) A body covers 100 m in 20 s. Its speed is:
 - (A) 5 m/s
 - (B) 20 m/s
 - (C) 0.2 m/s
 - (D) 2 m/s
- (vii) A car of speed is 5 km/h. what is the speed in m/s?
- (viii) An object is sliding down an inclined plane. The velocity changes at a constant rate from 10 m/s to 15m/s in two seconds. What is its acceleration?
- (ix) A bus accelerates uniformly from rest and acquires a speed of 36 km/h in 10 seconds. The acceleration is:
- (x) A person is moving along a rectangular path (ABCD), where the length of side AB is 12 m and of BC is 5 m. He starts from point A and goes to C via B. Calculate distance and displacement. Also find the speed if they cover distance in 2 sec.

7. PROBLEM SOLVING

Examples

(i) **If a body loses 2000 electrons, find the total charge on the body.**

Solution: When a body loses electrons, it becomes positively charged.

$$n = 2000, e = 1.6 \times 10^{-19} \text{C}$$

$$Q = n \times e = 2000 \times 1.6 \times 10^{-19} = 3.2 \times 10^{-16} \text{C}$$

(ii) **A conductor has 5×10^{20} electrons flowing in 20 seconds. Calculate the current.**

Solution: $n = 5 \times 10^{20}, t = 20 \text{ s}$

$$Q = n \times e = 5 \times 10^{20} \times 1.6 \times 10^{-19} = 80 \text{C}$$

$$I = \frac{Q}{t} = \frac{80}{20} = 4 \text{ A}$$

(iii) **Write three devices which work on heating effect of current.**

Answer: The devices which are working on heating effect of current are given below:

Bulb, Iron, Heater

(a) Electric bulb - tungsten glows

(b) Electric iron - nichrome heats

(c) Electric heater - heating element

(iv) **Why no magnetic monopoles exist?**

Answer: Magnetic monopoles (a single north or a single south pole) don't exist because magnetic field lines are always closed loops. They never start or end at a point.

Every magnet has a north and a south pole. If we cut a magnet in half, we don't get an isolated north or south pole we get two smaller magnets, each with both poles. This keeps happening no matter how much we cut it.

(v) **Write two uses of electromagnets.**

Answer: (a) Lifting heavy loads in junkyards.

(b) Removing magnetic particles from eye.

(vi) **A ray of light strikes a plane mirror such that the angle between the incident ray and the mirror surface is 30° . Find the angle of incidence.**

Solution: The angle of incidence is the angle between the incident ray and the normal (which is 90° to the surface).

$$\text{Angle of Incidence } (\angle i) = 90^\circ - 30^\circ = 60^\circ.$$

(vii) **A boy runs towards a plane mirror with a speed of 2 m/s. With what speed does his image approach him?**

Solution: Relative speed of image with respect to object = $2v$.

$$\text{Speed} = 2 \times 2 = 4 \text{ m/s.}$$

- (viii) **An object is placed 4 m in front of a plane mirror. How far is the image from the mirror? And what is the total distance between the object and its image?**

Solution: In a plane mirror, Object Distance (u) = Image Distance (v).

Image distance = 4 m.

Total Distance = Object Distance + Image Distance

Total Distance = 4 + 4 = 8 m.

- (ix) **A car moving at 30 m/s applies brakes and stops in 5 seconds. Find acceleration.**

Solution: Given: $u = 30$, $v = 0$, $t = 5$.

Then,

$$a = \frac{0 - 30}{5} = -6 \text{ m/s}^2$$

(Negative acceleration is retardation).

- (x) **A train travels the first 100 km in 2 hours and the next 100 km in 3 hours. Find the average speed.**

Solution: Total Distance = 100 + 100 = 200 km.

Total Time = 2 + 3 = 5 hours.

$$\text{Average Speed} = \frac{200}{5} = 40 \text{ km/h}$$



Practice Questions - 7

- (i) Which does NOT conduct current?
 - (A) Copper
 - (B) Rubber
 - (C) Graphite
 - (D) Sea water
- (ii) When 2 C flows in 4 s, current is:
 - (A) 8 A
 - (B) 2 A
 - (C) 0.5 A
 - (D) 6 A
- (iii) Component allowing/stopping current:
 - (A) Battery
 - (B) Bulb
 - (C) Switch
 - (D) Wire
- (iv) No current flows in:
 - (A) Closed circuit
 - (B) Open circuit
 - (C) Series circuit
 - (D) Complete circuit
- (v) Electromagnet loses magnetism when:
 - (A) Current increases
 - (B) Current off
 - (C) More turns
 - (D) Iron core added
- (vi) Magnet cut in 5, total poles:
 - (A) 2
 - (B) 5
 - (C) 10
 - (D) 15
- (vii) If an object is 5 cm high, the size of its image in a plane mirror is:
 - (A) 10 cm
 - (B) 2.5 cm
 - (C) 5 cm
 - (D) Depends on distance
- (viii) Bouncing back of light from a smooth surface is called:
 - (A) Absorption
 - (B) Reflection
 - (C) Bending
 - (D) Passing
- (ix) Which mirror can form a Real image?
 - (A) Plane Mirror
 - (B) Convex Mirror
 - (C) Concave Mirror
 - (D) Shaving Mirror
- (x) A body starting from rest has an initial velocity (u) of:
 - (A) 10 m/s
 - (B) 100 m/s
 - (C) 0
 - (D) Variable

(xi) Very Short Answer Type Questions

- (a) Define electric circuit.
- (b) What is the acceleration of a car moving with a constant velocity?
- (c) What is the value of the speed of light in air?
- (d) Convert 72 km/h into m/s.
- (e) Name the mirror used in torches and searchlights.

(xii) Short Answer Type Questions

- (a) Distinguish between Convergent and Divergent beams of light.
- (b) Explain Uniform Velocity and Non-Uniform Velocity.
- (c) Differentiate between Real and Virtual images (2 points).

(xiii) Long Answer Type Questions

- (I)** (a) Explain electric cell structure, working. Differentiate primary and secondary with examples.
- (b) Draw circuit (battery 3 cells, bulb, switch). Explain ON/OFF. Explain heating effect and applications.
- (II)** (a) A bus decreases its speed from 80 km/h to 44 km/h in 5 seconds. Find the acceleration (in m/s^2) of the bus.
- (b) A person travels along the boundary of a square field of side 10m. He completes one round in 40s. Calculate the distance (d) and magnitude of displacement (s) at the end of 2 rounds.

Practice Questions - Answer Key

(1) (i) (B)	(ii) (B)	(iii) (B)	(iv) (C)
(v) (C)	(vi) (C)	(vii) 3	(x) $k = 2$
(2) (i) (B)	(ii) (D)	(iii) (C)	(iv) (C)
(v) (B)	(vi) (C)		
(3) (i) (B)	(ii) (C)	(iii) (C)	(iv) (A)
(v) (D)	(vi) (C)		
(4) (i) (B)	(ii) (C)	(iii) (A)	(iv) (A)
(v) (C)	(vi) (C)	(vii) 50°	(viii) 130°
(5) (i) (B)	(ii) (B)	(iii) (C)	(iv) (B)
(v) (C)	(vi) (C)	(vii) 30 cm	(viii) 15 m/s
(6) (i) (D)	(ii) (C)	(iii) (C)	(iv) (B)
(v) (A)	(vi) (A)	(vii) 1.38 m/s	(viii) 2.5 m/s^2
(ix) 1 m/s^2	(x) Distance = 17 m Displacement = 13 m Speed = 8.5 m/s		
(7) (i) (B)	(ii) (C)	(iii) (C)	(iv) (B)
(v) (B)	(vi) (C)	(vii) (C)	(viii) (B)
(ix) (C)	(x) (C)	(xi) (d) 20	(xiii) (II) (a) -2 (b) $d = 80 \text{ m}$ $s = 0 \text{ m}$